# TOPS Technologies

## Interview Questions — Data Vizualization with Power BI

**Total Questions:** 57

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**1.** What is Power BI and what are its main components?

> **Hint:** Think about the tools you use to create, view, and share reports.

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**2.** How does the workflow in Power BI move from raw data to published dashboards?

> **Hint:** Consider the steps: data, transform, model, visualize, publish.

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**3.** What is the difference between a Power BI Dashboard and an Excel Dashboard?

> **Hint:** Focus on interactivity, data sources, and sharing capabilities.

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**4.** After installing Power BI Desktop, which pane would you use to add a new visualization to your report?

> **Hint:** Look for the pane that contains different chart types.

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**5.** Name and briefly describe the three main views in Power BI Desktop.

> **Hint:** Each view is used for a different part of the report-building process.

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**6.** What is the purpose of the Fields pane in Power BI Desktop?

> **Hint:** This pane lists all your tables and columns.

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**7.** Explain the difference between Import and DirectQuery when connecting to data sources in Power BI.

> **Hint:** Think about where the data is stored and how often it updates.

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**8.** How would you connect Power BI to both an Excel file and a web data source in the same report?

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**9.** What does setting Data Refresh in Power BI allow you to do?

> **Hint:** Consider how your report stays up-to-date with new data.

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**10.** In Power Query Editor, how do you remove all rows with null values in a specific column?

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**11.** What steps would you follow to remove duplicate rows in Power Query Editor?

> **Hint:** Think about transforming data before loading it into your report.

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**12.** How can you split a column into two in Power Query Editor?

> **Hint:** Look for options related to delimiters or fixed width.

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**13.** Describe how you would merge two columns into one in Power Query Editor.

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**14.** If you need to change a column's data type from text to number, where would you do this in Power BI?

> **Hint:** This is done during data transformation, not visualization.

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**15.** What does the 'Fill Down' operation do in Power Query Editor?

> **Hint:** Think about how missing values can be filled using values above.

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**16.** Explain the difference between 'Unpivot Columns' and 'Pivot Columns' in Power Query Editor.

> **Hint:** One turns columns into rows, the other does the reverse.

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**17.** How would you use the 'Group By' feature to get the total sales per region in your data?

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**18.** What is the purpose of creating a Custom Column in Power Query Editor?

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**19.** Describe a scenario where you would use a Conditional Column in Power Query Editor.

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**20.** What is the difference between merging queries and appending queries in Power BI?

> **Hint:** Consider whether you're joining tables side-by-side or stacking them vertically.

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**21.** How would you combine multiple CSV files from a folder into a single table in Power BI?

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**22.** What does the Query Dependencies view help you visualize in Power BI?

> **Hint:** It shows relationships between queries and data sources.

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**23.** Explain the difference between a Star Schema and a Snowflake Schema in data modeling.

> **Hint:** Focus on table structure and normalization.

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**24.** What is a one-to-many relationship in Power BI, and why is it important?

> **Hint:** Think about how tables are linked for analysis.

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**25.** How do you identify the primary key and foreign key in your data model?

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**26.** What is the difference between calculated columns and measures in Power BI?

> **Hint:** Consider when each is evaluated and how they're used in visuals.

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**27.** How would you create a hierarchy for a date field in Power BI?

> **Hint:** Think about grouping by year, quarter, month, etc.

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**28.** What is the purpose of setting a field as 'hidden' in your data model?

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**29.** Name three basic visuals you can use in Power BI and describe when you might use each.

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**30.** How does a Matrix visual differ from a Table visual in Power BI?

> **Hint:** One allows for row and column grouping.

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**31.** What is a KPI visual used for in Power BI?

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**32.** When would you choose to use a Line Chart instead of a Bar Chart in Power BI?

> **Hint:** Think about the type of data each visual is best for.

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**33.** Describe a scenario where a Combo Chart would be more effective than a single chart type.

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**34.** What is the purpose of a Waterfall chart in Power BI?

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**35.** How can you use a Map visual in Power BI, and what data do you need for it?

> **Hint:** Consider geographical fields like city or country.

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**36.** What is drill-down in Power BI visuals, and how does it help users explore data?

> **Hint:** Think about navigating from summary to detail.

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**37.** How would you apply conditional formatting to a visual in Power BI?

> **Hint:** Look for options to change colors based on values.

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**38.** What are custom visuals in Power BI, and how do you add them to your report?

**39.** What is DAX and where is it used in Power BI?

**40.** Explain the difference between row context and filter context in DAX.

> **Hint:** Consider how calculations are evaluated across rows and filters.

**41.** Write a simple DAX measure to calculate the total sales from a 'Sales' column.

**42.** How would you use the CALCULATE() function to apply a filter to a measure in Power BI?

> **Hint:** Think about modifying the filter context for your calculation.

**43.** What is the purpose of the VAR...RETURN syntax in DAX?

> **Hint:** It helps you store intermediate values in calculations.

**44.** How would you create a calculated column to extract the year from a date field using DAX?

**45.** Explain how the SAMEPERIODLASTYEAR function is useful for time-based analysis.

**46.** What is the difference between Page, Visual, and Report level filters in Power BI?

> **Hint:** Consider the scope of each filter type.

**47.** How do slicers help users interact with Power BI reports?

**48.** What is the purpose of bookmarks and buttons in Power BI navigation?

> **Hint:** These features help create interactive report experiences.

**49.** How would you use Edit Interactions to control how visuals affect each other in a report?

**50.** What steps would you take to design a consistent theme and layout for your Power BI report?

**51.** Explain what Row Level Security (RLS) is and why it is important when sharing Power BI reports.

> **Hint:** Think about restricting data access based on user roles.

**52.** Describe the process for publishing a Power BI report to the Power BI Service and sharing it with others.

**53.** How could you use an AI tool like ChatGPT to help you write a DAX formula or troubleshoot a Power BI error, and what would you do to ensure the AI's suggestion is correct?

> **Hint:** Think about verifying AI-generated code within your own report and understanding the logic.

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**54.** How can AI-powered features in Power BI, such as Q&A visuals, benefit business users, and where might you still need to rely on manual analysis?

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**55.** Suppose you have combined data from multiple sources, created relationships, and built visuals, but your numbers don't add up as expected. Walk through the steps you would take to troubleshoot this issue.

> **Hint:** Consider checking data refresh, relationships, filters, and DAX calculations.

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**56.** Integrate your knowledge: How would you use Power Query Editor, data modeling, and DAX together to prepare and analyze sales data from multiple regions?

> **Hint:** Think about cleaning data, creating relationships, and writing measures.

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**57.** Imagine you need to design an interactive sales dashboard that allows users to drill down from total sales to region, then to individual store. Which Power BI features would you use, and how would you set up the navigation?

> **Hint:** Consider hierarchies, drill-down, bookmarks, and slicers.

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